

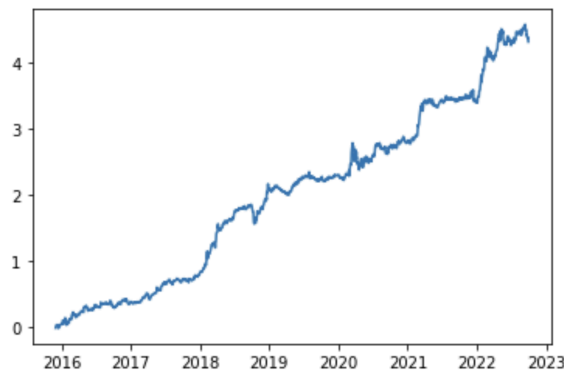
Overview

I developed and deployed a seemingly promising trading algorithm with a successful backtest just to witness its performance shift into decline. Although discouraging, this is a learning opportunity briefly discussed below.

The Strategy

Key details (specifics withheld) at the time of strategy creation at end of 2022:

1. The graph of cumulative sum of trade PL's is *upward-trending*:



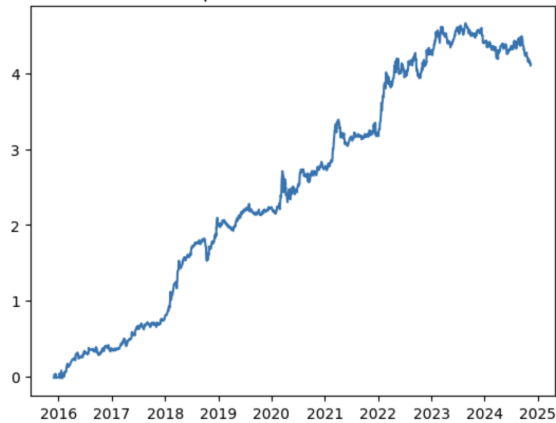
- a.
- b. The traded asset (undisclosed) price is primarily *horizontal-trending*, so it has *low correlation* with this upward-trending strategy
2. Profitable backtest performance metrics
 - a. 60% annualized return across 7 years (2016-2022)
 - b. 60% win rate at the trade-level
3. The strategy signal is built on *minutely* price data
 - a. The long/short signal is based on calculation comparisons on recent price history at a certain time. So, there is only 1 continuous parameter — the comparison time.
 - b. The backtest has 1,600+ trades (statistically significant)
 - c. The strategy was not found to work for other selected assets besides this one particular asset
4. 40% max historical drawdown (a little high)



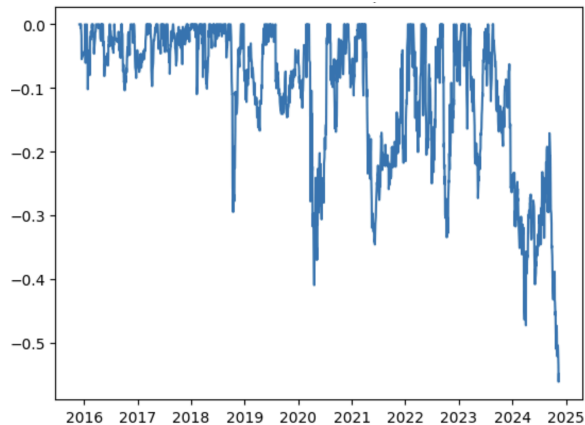
a.

The Downfall

At the end of 2024, it was identified visually from the cumulative performance graph that the strategy is now faltering — flattening in 2023 and then down-trending in 2024:



The drawdown is now unprecedented, exceeding -50%:



This is perhaps a great example of “past performance does not guarantee future performance” even after ensuring low correlation with the traded asset and ensuring a statistically significant 1,000+ trade count backtest history.

I have not observed anything fundamentally different with the traded underlying asset 2023 onwards, so it is not known why the strategy is “no longer working.” Could it work again one day? Maybe, maybe not. There can be many theories and explanations, but for now, there has been a fundamental shift in what used to make the strategy successful.